

Assignment 4: *Topic: E. 9.12*

Step 1: Problem Statement

Create a digital model of a FedEx Like Cargo terminal. This terminal will have 1 loading dock for trucks and one tarmac for planes. The truck and plane objects will be stored in arrays that will be sized and populated from two text files containing their respective details. The end goal will be to manipulate the arrays and an ArrayList that will contain reporting info, in order to sort the information on specific criteria (Truck Destination & plane capacity), format that info, and display it in an aesthetically pleasing way.

Step 2: What I know

I know how to create File objects, Scanner objects, Arrays, ArrayLists, classes, Objects, and How to populate the data.

I'm good with the concept of Has-A relationships.

Pretty good with String Formatting and creating the methods needed for the project.

Step 2: what I don't know

will need to do some work
compareTo method and using it
to sort my info. It seems
intuitive but I don't recall
having done it this way before

~~Step 3: Pseudo code~~ (main print terminal st

~~create new file objects for~~

~~myFile1 = FedExTrucks.txt~~

~~myFile2 = FedExPlaces.txt~~

~~create new Scanner to read Array Size~~

~~Scanner myFile1 = new Scanner~~

~~deckArray [myFile1.nextLine()]~~

~~TruckArray [myFile2.nextLine()]~~

~~// Populate truck array~~

~~while myFile1.hasNext~~

~~DeckName = myFile1.nextLine()~~

~~TruckName = myFile1.nextLine()~~

~~TruckDes = myFile1.nextLine()~~

~~next~~

Contains

Contains

Place
txt

Terminal

Farmac
class

Place Array

Deck
class

Truck Array

Truck
txt

Truck
class

place
class

Print
status
method

<arraylist>

Step 30 Pseudo code (main)

// create file objects & scanner
to read number docks & stacks
from files

new File myFile1 = FedExTrucks.txt;

new File myFile2 = FedExPlanes.txt;

close files;

new Scanner myScaw

numberDocks = myFile1.nextInt();

numberStacks = myFile2.nextInt();

// Call CargoTerminal to create ~~obj~~
terminal & populate Arrays with obj's

cargoTerminal(numberDocks, numberStacks);

~~// call addShipout & add CargoPlane
to populate Docking Deck and
Terminal~~

// call display CargoTerminal
to display 1st out put

cargoTerminal.displayCargoTerminal();

// call print TerminalStatus to print
sorted / Formatted info

printTerminalStatus(cargoTerm1);

Step 3 (cont): Pseudocode for PrintTermStatus

Access SemiArray & populate
arrayList<> with trucks

Collections.sort() by destination

Display toString() in Semi truck

Access planeArray & populate
arrayList<> with planes

Collections.sort() by capacity

Display toString in CarPlane

Step 4: What I Learned

I learned how to override
compareTo. Didn't seem like
it would be that hard but
took some work to get it
to function as expected.

Also feel like I learned about
"Has-A" relationships but feel
like there is a lot more to learn
~~on~~ that subject.

Lastly learned a little more about
String formatting & need to learn more.